

Library Management System

Overview

The Library Management System is a Java-based console application that helps manage library operations efficiently. The project uses Java, JDBC, and MySQL to store and manage book records. It follows Object-Oriented Programming (OOP) principles and provides a menu-driven interface for performing various library management tasks.

Features

Book Management

- Add new books
- View all books
- Search books by ID
- Update book details
- Delete books

Library Operations

- Issue books to users
- Return issued books
- Track issued book status

Database Integration

- MySQL database connectivity using JDBC
- Persistent storage of book records
- SQL-based CRUD operations

User Interface

- Menu-driven console application
- Interactive user input using Scanner
- Easy-to-use command-line interface

Technologies Used

- Java
- JDBC (Java Database Connectivity)
- MySQL
- MySQL Connector/J
- VS Code

Project Structure

```
Library-Management-System
|
├── src
|   ├── dao
|   │   └── BookDAO.java
|   ├── model
|   │   ├── Book.java
|   │   ├── User.java
|   │   └── Transaction.java
|   ├── service
|   │   └── Library.java
|   ├── util
|   │   ├── DBConnection.java
|   │   └── DataManager.java
|   └── LibraryApp.java
├── LIB
|   └── mysql-connector-j-9.7.0.jar
└── README.md
```

Database Schema

Books Table

```
CREATE TABLE books (
    id INT PRIMARY KEY,
    title VARCHAR(255),
    author VARCHAR(255),
    issued BOOLEAN DEFAULT FALSE,
    issued_to VARCHAR(100),
    issue_date DATE
);
```

Setup Instructions

1. Clone Repository

```
git clone <repository-url>
cd Library-Management-System
```

2. Configure MySQL

Create a database:

```
CREATE DATABASE library_db;
USE library_db;
```

Create the books table using the schema provided above.

3. Configure Database Connection

Update the database credentials in `DBConnection.java`:

```
private static final String URL = "jdbc:mysql://localhost:3306/library_db";
private static final String USER = "root";
private static final String PASSWORD = "your_password";
```

4. Add MySQL Connector

Download MySQL Connector/J and place the JAR file inside the `LIB` folder.

5. Compile

```
javac -cp "LIB/mysql-connector-j-9.7.0.jar;src" -d src src/LibraryApp.java
```

6. Run

```
java -cp "LIB/mysql-connector-j-9.7.0.jar;src" LibraryApp
```

Learning Outcomes

Through this project, the following concepts were implemented and practiced:

- Object-Oriented Programming (OOP)
- JDBC Database Connectivity
- SQL CRUD Operations
- Exception Handling
- Collections and Data Management
- Layered Architecture (DAO, Model, Utility)
- MySQL Database Design
- Console-Based Application Development

Future Enhancements

- User Management Module
- Authentication System
- Fine Calculation for Late Returns
- Transaction History Tracking
- Dashboard and Analytics
- GUI using JavaFX or Swing
- REST API Integration

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This project is developed for educational and learning purposes.